

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mechanical Engineering) SEMESTER - 4 Summer 2025 (Regular)

Course : Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTMC401 - MANUFACTURING PROCESSES - I

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 In sand casting, the main function of the core is to:
A) Strengthen the mould
B) Make hollow sections in casting
C) Improve surface finish
D) Allow easy flow of molten metal
- Which one of the following is a permanent mould casting process?
A) Investment casting
B) Die casting
C) Shell moulding
D) Sand casting
- 3 The main advantage of cold rolling over hot rolling is:
A) Better surface finish
B) Higher ductility
C) Less force required
D) Increased grain size
- 4 Cold forging is generally used when:
A) Material is brittle
B) Material is soft and ductile
C) High precision and good surface finish is required
D) No die is available
- Extrusion is a process in which:
A) Material is compressed in a die to reduce thickness
B) Material is drawn through a die to reduce diameter
C) Material is forced through a die to produce desired cross-section
D) Material is hammered into a mould
- 6 A key limitation of wire drawing is:
A) Poor surface finish
B) Limited to high temperatures only
C) Limited to ductile materials
D) Requires casting before drawing
- 7 In shielded metal arc welding (SMAW), the electrode is:
A) Non-consumable
B) Made of copper
C) Used only as filler
D) Consumable and coated with flux

- 8 Weld porosity is usually caused by:
 A) Excessive filler metal B) Slow welding speed
 C) Entrapped gases D) High arc voltage
- 9 A reamer is used for:
 A) Making a hole B) Producing internal threads
 C) Finishing a drilled hole to accurate size D) Chamfering the edges

- Tapping operation is generally performed after:
 A) Facing B) Reaming
 C) Drilling D) Boring
- In broaching, the metal is removed by:
 A) One cutting edge only B) Multiple cutting edges progressively
 C) Rotating the workpiece D) Heat and pressure
- 12 Which milling operation is best suited for cutting internal threads?
 A) Thread milling B) Form milling
 C) End milling D) Profile milling

Q2. Solve the following.

- A) Explain in detail the various steps involved in making sand castings. 6
 B) Explain the centrifugal casting with the help of neat sketch. 6

Q3. Solve the following.

- A) Enlist the different types of rolling processes. Explain the continuous rolling with the help of neat sketch. 6
 B) Explain closed die forging with the help of neat sketch. 6

Q4. Solve Any Two of the following.

- A) Classify the extrusion processes and, differentiate between hot and cold extrusion. 6
 B) Explain the hydrostatic extrusion with the help of neat sketch. 6
 C) Explain the working principle of wire drawing process with the help of neat sketch. 6

Q5. Solve Any Two of the following.

- A) Describe the working principle of Tungsten Inert Gas (TIG) welding. What are its advantages over consumable electrode welding? 6
 B) Explain the Submerged Arc Welding process with a neat sketch. Why is this process considered to be highly productive? 6
 C) Define machining. Enlist the different types of machining processes and explain its importance in manufacturing. 6

Q6. Solve Any Two of the following.

- A) Describe the main parts of a lathe machine with the help of a labeled diagram. 6
 B) What is reaming? How does it differ from drilling and boring? 6
 C) What is broaching? Discuss internal and external broaching with neat diagrams. 6